



Fostering collaborative futures: Multidisciplinary approaches to AI integration in educational ecosystems

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ABSTRACT

With the rapid evolution of AI and data-driven technologies, AI integration in the educational ecosystem requires careful consideration. This paper aims to facilitate the process of fostering collaborative futures with AI agents in education by examining the impact of AI on education from the perspectives of researchers, practitioners, and policymakers. The authors conducted a Delphi study involving a survey administered to $N = 70$ international professionals, followed by in-depth face-to-face discussions with a panel of international experts from policy, research, and practice to identify key trends and challenges for deploying AI in education. The results indicate that the three most important trends were (1) AI-related competences for learners and educators, (2) automated formative assessments and feedback with AI agents, and (3) the evolved roles of teachers to teach with AI agents. Additionally, top importance included (4) the ongoing imperative to explainability and trustworthiness of AI agents and (5) accessibility and equal access to AI agents. Based on these findings, the paper outlines policy and action recommendations for collaborative AI initiatives in education and proposes a research agenda to address identified research gaps.

Introduction

Artificial intelligence (AI) in education represents a broad term that encompasses a diverse range of technologies, algorithms, and related multimodal data used to support teaching and learning in formal, non-formal, and informal contexts. As AI continues to evolve, its integration into educational ecosystems prompts significant opportunities and challenges [1,2]. An education ecosystem is a dynamic, sustainable, and interconnected network of individuals, institutions, data, technologies, policies, and processes, as well as cultures, that collectively influence and impact teaching and learning in formal, non-formal, and informal contexts [3]. The education ecosystem approach emphasises the need for AI to be designed with pedagogical considerations in mind, as well as its ability to adapt and respond dynamically to educators' and learners'

needs based on real-time data analysis [4,5]. Hence, the integration of AI within educational ecosystems necessitates a careful examination of ethical implications, including issues related to data privacy, bias in algorithmic decision-making, and the potential for widening existing educational inequalities [6,7]. Although prior studies increasingly recognized the importance of multiple stakeholders —such as teachers, students, researchers, policymakers, and industry partners [8]— many existing studies still examined these actors, issues, or technologies in isolation focusing separately on teacher competencies [9], student use of AI tools [10] or policy and ethics [11] without sufficiently examining how these dimensions interact within a broader educational ecosystem.

This paper is based on (a) a literature review focussing on the impact of artificial intelligence in the context of education, (b) a Delphi study [12] involving $N = 70$ international experts and a persona-based

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focus-discussion on current opportunities and challenges of AI in relation to educational ecosystems, resulting in outlining policy recommendations and a research agenda for closing identified research gaps. The trends which emerged from the Delphi study include (1) AI-related competences for learners and educators; (2) Automated assessment and feedback with AI agents; (3) Evolved roles of teachers while teaching with AI agents; (4) Explainability and trustworthiness of AI agents; and (5) Accessibility and equal access of AI agents.

AI adoption in education is not solely a technological issue, but rather a systemic process shaped by the interrelationships among pedagogical practices, institutional structures, governance, infrastructure, stakeholder capacities, and equity considerations. Novelty of this study lies in its ecosystem perspective, which aimed to provide a more integrated understanding of how these dimensions influence one another and collectively shape future directions for AI in education. The paper will elaborate on the methodology and findings of the Delphi study, as well as provide recommendations for policymakers (PM), researchers (RE), and practitioners (PR), each strategy linked to the corresponding trends identified above.

Background

Artificial intelligence in education

The integration of Artificial Intelligence (AI) in education has evolved significantly, transitioning from early intelligent tutoring systems to sophisticated generative AI tools with the potential to personalise, automate, and transform teaching and learning. Historically, AI in education (AIED) has focused on adaptive learning systems, automated feedback, and intelligent agents, but recent developments such as large language models and multimodal data processing have broadened the scope of possibilities [13,14]. These advancements facilitate diverse applications, from personalised learning environments and real-time feedback systems to AI-supported content creation and virtual teaching assistants.

Systematic reviews have identified five principal domains within AIED: personalised learning, AI-supported formative assessment, virtual tutors and learning companions, evolving teacher roles, and ethical governance of AI systems [13,14]. Pursuing the benefits of AI in education is vital as it indicates large effect sizes on students' learning achievement [15]. However, the literature also underscores several pressing concerns. These include limited transparency in AI algorithms [16], a lack of alignment with pedagogical frameworks and clear educational standards [2,17,18], as well as challenges in addressing algorithmic bias and fairness [6]. Several studies (e.g. Casal-Otero et al. [9], Delcker et al. [19] and Su et al. [17,18]) further note the urgency of developing AI-related competencies for both learners and educators to ensure effective and critical engagement with AI systems.

The emergence of generative AI tools such as ChatGPT has intensified debates around assessment, academic integrity, and the shifting boundaries of human-machine collaboration in education [20]. While AI presents enormous opportunities, researchers stress that its integration must remain human-centred, pedagogically sound, and ethically governed [21]. The growing consensus is that successful AI implementation requires a broader ecosystemic perspective, where infrastructure, policy, pedagogy, and ethical practices co-evolve and are interdependent. This sets the stage for a deeper examination of the data and infrastructural needs that underpin such ecosystems.

Data needs in education

AI-enhanced educational ecosystems are underpinned by robust data infrastructures that enable AI systems to interpret, predict, and adapt to learner and educator needs. This shift towards data-intensive practices, a phenomenon referred to as the datafication of education, has transformed how learning processes are recorded, analysed, and acted upon

[22]. Data needs in education, therefore, encompass the availability, accuracy, quality, interoperability, governance, and ethical use of data generated through digital learning environments.

The literature identifies learning analytics and educational data mining as key domains where AI intersects with data to support personalisation, institutional decision-making, and performance prediction. Yet, the effectiveness of such systems depends heavily on the accuracy, completeness, and representativeness of the underlying data [21]. Biases embedded in datasets, whether due to socio-economic, geographic, or demographic imbalances, can reinforce educational inequalities [6]. As such, attention to algorithmic fairness, data representativeness, and transparent decision-making processes is critical [11].

Privacy and data protection also feature prominently [23]. Students and educators express growing concern over opaque data practices, surveillance risks, and lack of control over personal information [24,25]. Regulatory frameworks such as the EU's General Data Protection Regulation (GDPR) and UNESCO's ethical guidelines advocate greater accountability, informed consent, and learner agency in data governance. However, research highlights the need for improved data literacy among stakeholders, alongside technical solutions for data de-identification and secure sharing.

Equitable data infrastructures, those that support interoperability, open standards, and inclusive participation, are essential to realising the benefits of AI. This requires investment in both technological systems and human capabilities. Ultimately, meeting the data needs of AI-enhanced education is not merely a technical endeavour, but a socio-technical one, demanding holistic, values-driven approaches [26]. These considerations are integral to the concept of educational ecosystems explored in the following section.

Educational ecosystem

As a working group, we defined an education ecosystem as a dynamic, sustainable, and interconnected network of individuals, institutions, data, technologies, policies, and processes, as well as cultures that collectively influence and impact the teaching and learning in formal, non-formal, and informal contexts. To understand the increased complexity, an ecosystem perspective [27,28] is often considered. While studies on ICT integration and eLearning adoption are conducted across various geopolitical and cultural contexts (e.g., Brazil; [29]; Europe; [30,31]), there is a pressing need to focus on AI and data-driven services within the current educational ecosystem.

Context and research questions

EDUsumMIT is an UNESCO (United Nations Educational, Scientific and Cultural Organization; <https://www.unesco.org>)-endorsed global community of researchers, policy-makers, and practitioners committed to supporting the effective integration of Information Technology (IT) in education by promoting active dissemination and use of research. Approximately 130 leading researchers, policymakers, and practitioners from all continents and across 20 countries gathered in Dublin, Ireland, from June 23 to 25, 2025, to discuss emerging themes and define corresponding action items. Prior to the meeting, thematic working groups (TWGs) conducted research related to current challenges in educational technologies with a global impact. This paper is based on the work of the TWG focusing on 'Artificial Intelligence (AI), big data, teaching and learning: future challenges and opportunities'. The authors of this article constituted the TWG, which addressed the following research questions:

1. What are recent research and innovations in artificial intelligence linked to supporting learning, teaching, and educational decision-making?
2. What recommendations for artificial intelligence integration in education can be proposed for policy, practice, and research?

Delphi study

This study aimed to uncover global trends and educational practices pertaining to AI in education. A panel of multinational specialists from industry and research institutions reached a consensus on a set of current trends using the Delphi method. The process involved the following four phases: Phase 1 aimed to identify topic trends. It was achieved by 24 May 2025. Phase 2 aimed to expand the expert base and synthesise topic trends. Phase 3 conducted a literature review related to the final list of trends. Phase 4 focused on preparing the report for publication and outreach.

Methodology

The Delphi method is a robust approach for determining forecasts or policy positions considered essential [12]. A Delphi study can be conducted using paper-and-pencil instruments, computer- or web-based approaches, as well as face-to-face communication processes. For this study, the researchers applied a mixed Delphi design, including (a) computer-based and (b) face-to-face discussion methods.

In the first phase, using the *computer-based method*, a panel of international researchers in artificial intelligence in education was invited to submit trends and institutional practices related to AI in the educational arena. The initial list of trends was then aggregated through consensus, resulting in a final, consolidated list of 15 topics related to AI in education. In an additional step of the computer-based method, the list was disseminated to global specialists in AI in education. Each participant was asked to rate the 15 topics on the list concerning (1) importance, (2) impact, and (3) key challenges on a scale of 1 – 10 (with 10 being the highest). The instructions for the ratings were as follows:

- Please rate the **IMPORTANCE** of each of the trends (on a scale of 10, where 10 is the highest IMPORTANCE) for learning and teaching related to AI in organisations within the next three years.
- Please rate the **IMPACT** of each of the trends (on a scale of 10, where 10 is the highest IMPACT) on learning and teaching related to AI and how organisations will utilise them.
- Please rate the **KEY CHALLENGES** of each of the trends in AI in education (on a scale of 10, where 10 is the highest CHALLENGE) that organisations will face within the next three years.

In preparation for the second phase, *face-to-face discussion*, the panel of international researchers was asked to provide three relevant scientific literature resources related to the identified key areas in the first phase and explain their contribution to the respective area of development. Next, the panel of international researchers convened for a three-day face-to-face workshop. During the face-to-face meeting, the panel of international researchers and policymakers discussed and agreed on several trends, challenges, and recommendations concerning research gaps and important implications for educational stakeholders, including policymakers and practitioners.

We used persona-based focus-group discussions to deepen and contextualise the Delphi findings. User personas are fictional profiles rooted in real evidence, designed to represent the different attitudes and needs of a user base [32]. Prior to the workshop, members of the thematic working group compiled model-card style notes and field observations from cross-visits to other EDUsummIT working groups, in particular TWG 1 (emerging technologies in schools), TWG 4 (professional digital competence of teachers), and TWG 5 (AI literacy for teaching and learning). These notes captured stakeholder-specific concerns about AI, including datafication, ethical use of student data, professional learning, and translation between research, policy, practice and industry.

On the basis of these notes, we synthesised four personas that represented key stakeholders in AI-enhanced educational ecosystems:

- *The Teacher (Educator)*, who is primarily concerned with classroom implementation, workload and curriculum pressures, guidance on safe and effective AI use, and the implications of AI for assessment, student well-being and professional identity.
- *The Ed-tech Developer*, who focuses on product efficacy, access to anonymised and identifiable learner data, interoperability with learning management systems, and compliance with data protection and privacy regulations.
- *The Researcher*, who emphasises adherence to institutional and national ethics and privacy guidelines, the use of evidence-informed and rapid-cycle research to evaluate AI tools, and the need for sustainable research-practice-industry networks.
- *The Policymaker*, who prioritises funding decisions, regulatory frameworks, alignment with school improvement plans and curriculum standards, engagement with unions and other stakeholders, and clarity around acceptable uses of AI in instruction and assessment.

Workshop participants were assigned to small groups and asked to adopt one persona when discussing the ranked AI trends. From the viewpoint of their assigned persona, groups identified concrete needs, anticipated challenges and potential policy, practice and research recommendations. These discussions were documented through written notes and flip charts, then synthesised narratively. The resulting persona-based insights were used to refine our interpretation of the Delphi survey results and to inform the cross-stakeholder strategy and action recommendations reported in the later sections of this paper.

Participants

A total of $N = 70$ participants (36 female, 33 male; 1 not reported) with 41.43% in the 40–49 years age band took part in the study. The professional expertise of participants was distributed among academia ($n = 26$), research ($n = 23$), industry ($n = 5$), policy ($n = 8$), and teachers ($n = 8$), with an average of 19.5 years ($SD = 10.69$) of professional experience. The geographic distribution was as follows: Australia ($n = 3$), Croatia ($n = 1$), France ($n = 1$), Germany ($n = 15$), Greece ($n = 1$), India ($n = 2$), Ireland ($n = 16$), Israel ($n = 1$), Italy ($n = 2$), Japan ($n = 1$), Netherlands ($n = 7$), Philippines ($n = 1$), Singapore ($n = 4$), Sweden ($n = 2$), Taiwan ($n = 1$), United Kingdom ($n = 4$), USA ($n = 7$), Vietnam ($n = 1$). Fig. 1 presents the map of the participants.

Data analysis

All expert survey response data were saved and analysed using an anonymised process as per conventional research data protection procedures. Data were cleaned and combined for descriptive and inferential statistics using R Statistics (<https://www.r-project.org>). All effects were tested at the .05 significance level, and effect size measures were computed where relevant.

Result

Phase 1 – global trends in artificial intelligence in education

At the end of phase 1, we received and compiled a list of 56 trends from 17 experts (Compiled in Appendix 1). They were subsequently grouped into 15 themes (presented in Table 1).

Phase 2 – global trends in artificial intelligence in education

For the second phase, the top three trends in terms of importance, impact, and challenges of AI in education were critically reflected upon and linked to an in-depth, research-informed group discussion. However, all other trends have been recognised during the consensus phase to develop recommendations for strategies and actions.



Fig. 1. Geographic distribution of experts (N = 70) involved in this study.

Table 1
List of compiled themes and example trends mentioned by the experts (n = 17).

Sl. No.	Theme	Example trends
1	AI agents as Virtual Teaching Assistants	Virtual teaching assistants
2	AI-Enhanced Educational Analytics	AI-Enhanced Learning Analytics
3	AI-assisted Educational Content Creation	AI and/in Content Creation - potentials and problems for education.
4	AI for Educational Administrative Efficiency	Automation of Teacher Tasks.
5	AI Augmentation with Educational Edge Devices	Seamless Integration of AI with Augmented and Virtual Reality (AR/VR)
6	Human-AI Collaborative Models in Learning Processes	AI facilitated behavioural modification at a societal scale through education, mass, and social media
7	Interdisciplinary Curriculum through AI-Driven Knowledge Graphs	Optimising STEM-Humanities Interdisciplinary Curriculum through AI-Driven Knowledge Graphs
8	AI-Powered transformations: Data wrappers and middleware	AI-Powered Data Wrappers: Enhancing Institutional Data Insights without Disrupting Legacy Systems
9	Personalised and Adaptive learning with AI agents	Generative AI for personalised and adaptive learning (e.g., learning companions)
10	Automated formative assessment and feedback with AI agents	Automated assessment and feedback using LLMs
11	AI-related competences for learners and educators	Data Literacy for teachers and students in the age of AI
12	AI-assisted skill development	Using big data on career and job opportunities to craft multiple pathways for students
13	Evolved roles of teachers while teaching with AI agents	Generating research-based, evidence-proven instructional strategies for educators to use with students
14	Explainability and Trustworthiness of AI Agents	Trustworthiness and reliability of the algorithms, and the ability to justify decisions to learners and educators
15	Accessibility and Equal Access of AI agents	AI Integration in schools

Table 2 summarises the experts’ ratings of **Impact**, **Importance**, and **Challenges** across fifteen AI-in-education themes, ordered by mean Impact. The mean, standard deviation, and minimum score of the experts are presented. The impact and importance scores are colour-coded in a green gradient, and challenges are colour-coded in a red gradient (with higher values indicated by deeper colours). The maximum score was 10 for all themes, except for the Impact score of AI augmentation with educational edge devices, which was 9. A priority indicator was developed based on the difference between the average perceived importance and the perceived challenge of each theme. After sorting by impact scores, the priority indicators were reviewed for the top three themes in terms of both impact and importance.

The highest-rated theme in terms of **Impact** was *AI-related competences for learners and educators* (M = 8.57), which also emerged as the most **Important** (M = 8.64). Although it was perceived as relatively challenging (M = 7.49), it still held a positive priority indicator, signifying strong consensus that competence-building is foundational for AI adoption in education.

The second-highest Impact rating was shared by *Automated formative assessment and feedback with AI agents* (M = 8.13) and *Evolved roles of teachers while teaching with AI agents* (M = 8.13). However, their challenge profiles diverged: automated assessment showed moderate challenge (M = 7.13), while the evolved teacher role was rated one of the most challenging themes (M = 8.40), yielding a negative priority indicator. This suggests that redefining teacher responsibilities in AI-supported classrooms is perceived as highly complex despite its high impact.

Themes reflecting responsible and equitable AI—*Explainability and Trustworthiness of AI Agents* (Impact M = 7.84; Challenge M = 8.17) and *Accessibility and Equal Access to AI Agents* (Impact M = 7.81; Challenge M = 8.00)—also ranked prominently in Impact and Importance. Their high challenge scores indicate substantial barriers to implementation, reflecting ongoing concerns about ethics, infrastructure, and policy.

Other notable themes with relatively high priority indicators include *AI-assisted Educational Content Creation* (priority = 1.63), *AI for Educational Administrative Efficiency* (priority = 1.23), and *AI-related competences* (priority = 1.16). These themes combine high importance with comparatively lower levels of perceived challenge, making them strong candidates for near-term action.

Table 2

Impact, Importance, and Challenges of the themes. Ordered based on the mean score of Impact.

Themes	Impact Mean (SD, min)	Importance Mean (SD, min)	Challenge Mean (SD, min)	Priority indicator (importance - challenge)
AI-related competences for learners and educators	8.57 (1.92, 2.00)	8.64 (2.01, 1.00)	7.49 (2.12, 1.00)	1.16
Automated formative assessment and feedback with AI agents	8.13 (1.70, 2.00)	7.73 (1.99, 2.00)	7.13 (2.12, 1.00)	0.60
Evolved roles of teachers while teaching with AI agents	8.13 (2.04, 3.00)	7.91 (1.95, 2.00)	8.40 (1.81, 3.00)	-0.49
Personalised and Adaptive learning with AI agents	7.93 (1.88, 2.00)	7.80 (2.07, 1.00)	7.29 (1.79, 3.00)	0.51
Explainability and Trustworthiness of AI Agents	7.84 (2.15, 2.00)	8.39 (1.82, 3.00)	8.17 (2.18, 1.00)	0.21
Accessibility and Equal Access of AI agents	7.81 (2.44, 1.00)	8.19 (2.23, 2.00)	8.00 (2.26, 1.00)	0.19
AI-assisted skill development	7.79 (1.94, 2.00)	7.61 (1.93, 2.00)	7.04 (1.88, 3.00)	0.57
AI-assisted Educational Content Creation	7.76 (1.91, 2.00)	7.59 (1.87, 2.00)	5.96 (2.02, 1.00)	1.63
AI for Educational Administrative Efficiency	7.10 (2.13, 2.00)	7.10 (2.25, 2.00)	5.87 (2.19, 1.00)	1.23
AI agents as Virtual Teaching Assistants	7.01 (2.14, 2.00)	7.20 (2.16, 2.00)	6.76 (2.07, 1.00)	0.44
AI-Enhanced Educational Analytics	6.97 (2.38, 1.00)	7.14 (2.24, 1.00)	6.31 (2.07, 1.00)	0.83
Human-AI Collaborative Models in Learning Processes	6.77 (2.37, 1.00)	7.06 (2.39, 1.00)	6.89 (2.10, 1.00)	0.17
Interdisciplinary Curriculum through AI-Driven Knowledge Graphs	5.90 (2.57, 1.00)	5.96 (2.36, 1.00)	6.29 (1.99, 1.00)	-0.33
AI-Powered transformations: Data wrappers and middleware	5.60 (2.56, 1.00)	5.74 (2.47, 1.00)	5.93 (2.38, 1.00)	-0.19
AI Augmentation with Educational Edge Devices	5.39 (2.23, 1.00)	5.70 (2.22, 1.00)	5.87 (2.32, 1.00)	-0.17

Conversely, several lower-impact themes, such as *AI Augmentation with Educational Edge Devices* (Impact $M = 5.39$), *AI-powered Data Wrappers and Middleware* (Impact $M = 5.60$), and *Interdisciplinary Curriculum through AI-driven Knowledge Graphs* (Impact $M = 5.90$), also have negative priority indicators, suggesting both lower urgency and higher implementation complexity.

Overall, the importance ($M = 7.31$, $SD = .913$) of AI in education was rated higher than impact ($M = 7.24$, $SD = .976$) and challenge ($M = 6.89$, $SD = .859$), but they are not significantly different, $F(2, 42) = 0.91907$, $p > .05$.

Discussion

This Delphi study brought together global specialists from research institutions, industry, and policymaking to collaboratively examine emerging trajectories of AI integration within educational ecosystems. While the Delphi method is valuable for structuring expert dialogue and synthesising diverse viewpoints, it also carries inherent limitations. As noted by Beiderbeck et al. [33], the pursuit of consensus may

inadvertently narrow the scope of perspectives, potentially introducing bias into collective judgments. While participants from 18 countries participated, the current study also relied on a convenience sample, which may not fully reflect the breadth of expertise and varied experience levels in AI in education worldwide. Further we acknowledge the experts were practitioners, education administrators, researchers and policy makers who represented the stakeholders involved in designing and operationalizing policy decisions at the institutional level and in implementing AI-related tools in educational practice. Students and parents are also important stakeholders who are the target users of such developed systems. However they were not involved at this stage of the study. Future research would benefit from expanding the participant pool in each region to enhance representativeness and deepen the empirical grounding of the findings. Additionally, although some conceptual overlap emerged across the identified constructs, iterative face-to-face discussions among the international panel enabled continuous monitoring, refinement, and validation of the thematic distinctions. In addition, the Delphi approach and the persona-based discussions involved a degree of interpretive synthesis by the research

team. While this enabled the integration of diverse perspectives across stakeholder groups, it may also have shaped how themes were prioritised and articulated, potentially underrepresenting less dominant or dissenting viewpoints. As such, the findings should be interpreted as a structured expert-informed synthesis rather than a direct aggregation of individual perspectives.

Across the expert ratings, three categories of themes emerged that together illustrate the multifaceted priorities for advancing AI in education. First, foundational competence-building—particularly the development of AI-related competences for both learners and educators—was rated highest in impact and importance, underscoring the field's consensus that meaningful AI integration depends on human capacity and readiness. Second, themes related to pedagogical transformation, such as automated formative assessment and the evolving roles of teachers working alongside AI agents, were also viewed as highly impactful but posed notable challenges, particularly in reconfiguring teaching practices and professional identities. Finally, responsible and equitable AI implementation emerged as a critical area, with explainability, trustworthiness, and equal access to AI agents being rated as both essential and challenging to achieve. Collectively, these categories highlight that while AI holds significant promise for enhancing learning and teaching, its effective adoption requires parallel investments in human capability development, pedagogical redesign, and robust ethical and governance structures. The top trends are highlighted in Table 3 and discussed further.

These findings converge strongly with UNESCO's Guidance for Generative AI in Education and Research [34], which positions human capacity building, governance, transparency, and equity as central conditions for responsible adoption of generative AI in education. In particular, the highest-ranked Delphi theme, AI-related competences for learners and educators, aligns with UNESCO's emphasis on developing the knowledge and capabilities needed for critical and effective use. The prominence of evolving teacher roles similarly reflects UNESCO's insistence that AI should complement rather than displace teacher judgement and pedagogical responsibility. Likewise, the Delphi emphasis on explainability and trustworthiness corresponds with UNESCO's concern for transparency, accountability, and institutional oversight, while the prioritisation of accessibility and equal access reinforces UNESCO's focus on inclusion and equity. Where the present study extends this guidance is by showing that these are not only normative principles, but empirically prioritised concerns that experts perceive as high-impact and, in several cases, highly challenging to enact within educational ecosystems.

Trends identified for AI in education

• AI-related competences for learners and educators

As AI becomes increasingly integrated into educational environments, a critical need arises to cultivate specific competencies among learners and educators [9]. These competencies encompass technical skills and cognitive, ethical, and emotional dimensions, enabling effective engagement with AI technologies in education [19]. For learners, this involves understanding fundamental AI concepts such as machine learning algorithms, data processing, and the ethical implications associated with AI systems. Moreover, developing critical thinking skills is essential to evaluate AI-generated content critically, distinguishing between credible information and potential biases or inaccuracies inherent in AI outputs [35]. Educators face a parallel set of challenges. They must become proficient in utilising AI tools to enhance pedagogical practices while maintaining a critical perspective on their limitations [2].

• Automated formative assessment and feedback with AI agents

Automated assessment systems powered by AI agents are increasingly being deployed to provide real-time, individualised feedback that can significantly impact individual students' academic performance [36]. One of the primary opportunities of automated assessment is its ability to process vast amounts of data quickly and accurately [37]. Unlike traditional assessment methods that often provide delayed feedback after grading, AI-driven systems offer instantaneous insights into performance. This immediacy helps reinforce correct responses and rectify misunderstandings swiftly, thereby supporting a more iterative and reflective learning process [38]. However, the deployment of automated assessment systems raises several critical considerations. One significant challenge is ensuring the validity and reliability of AI-generated assessments and feedback [39].

• Evolved roles of teachers while teaching with AI agents

Traditionally seen as the primary source of knowledge and facilitators of learning, educators are now tasked with integrating AI technologies into their pedagogical practices, transforming their roles to meet contemporary educational demands [2]. This not only reshapes pedagogical practices but also raises questions about teacher agency and the balance between automated recommendations and an individual teacher's professional judgment. Teachers become orchestrators of more

Table 3
Top trends.

Themes	Impact Mean (sd, min)	Importance Mean (sd, min)	Challenge Mean (sd, min)	Priority indicator (importance - challenge)
AI-related competences for learners and educators	8.57 (1.92, 2.00)	8.64 (2.01, 1.00)	7.49 (2.12, 1.00)	1.16
Automated formative assessment and feedback with AI agents	8.13 (1.70, 2.00)	7.73 (1.99, 2.00)	7.13 (2.12, 1.00)	0.60
Evolved roles of teachers while teaching with AI agents	8.13 (2.04, 3)	7.91 (1.95, 2.00)	8.40 (1.81, 3)	-0.49
Explainability and Trustworthiness of AI agents	7.84 (2.15, 2.00)	8.39 (1.82, 3.00)	8.17 (2.18, 1.00)	0.21
Accessibility and Equal Access of AI agents	7.81 (2.44, 1)	8.19 (2.23, 2.00)	8.00 (2.26, 1.00)	0.19

student-centred learning environments that emphasise the development of learners' capacities to navigate complex information landscapes independently. Integrating AI into classrooms requires teachers to develop competencies in technology management and digital literacy [40], ensuring they understand the underlying mechanisms of AI tools to effectively incorporate them into curricula [10].

- Explainability and Trustworthiness of AI agents

Integrating AI into educational contexts raises critical questions about explainability and trustworthiness, which are essential to ensure that these technologies serve their intended purposes effectively and ethically. Explainability pertains to the transparency with which AI systems operate, enabling stakeholders (teachers, students, and administrators) to understand how decisions or recommendations are derived [41]. In education, where decisions can significantly impact learning outcomes, stakeholders must comprehend the rationale behind AI-generated insights or interventions. Without explainability, there exists a risk of eroding trust in these systems, as opaque algorithms might lead to decisions that appear arbitrary or unjustified [16]. Trustworthiness extends beyond mere transparency; it encompasses reliability, fairness, and accountability [42].

- Accessibility and Equal Access of AI agents

AI technologies are becoming increasingly prevalent in educational environments, and ensuring accessibility and equal access emerges as a paramount concern. Accessibility refers to the design and implementation of AI tools that are usable by all individuals, regardless of their physical or cognitive abilities [43]. Equal access extends beyond physical or cognitive accessibility to address broader socio-economic disparities that may limit some students' opportunities to benefit from AI technologies. Moreover, equitable access demands vigilance against algorithmic biases that could perpetuate existing inequalities within educational systems [6]. AI systems must be trained on diverse datasets representing a wide range of demographic groups to avoid reinforcing stereotypes or marginalising already disadvantaged populations.

A core contribution of this study is to position AI not as an isolated technological intervention but as a phenomenon embedded within and shaped by a broader educational ecosystem. The five trends identified in the Delphi study (AI-related competences, automated assessment, evolving teacher roles, explainability and trustworthiness, and equitable access) map directly onto the interdependent components of this ecosystem, including human actors, data flows, institutional infrastructures, cultural norms, and governance structures. Rather than treating these trends as discrete issues, an ecosystem perspective reveals how each trend functions as a node within a larger sociotechnical network: for example, competence development depends on institutional professional learning systems and policy support; responsible AI requires robust data governance and cultural norms of transparency; and equitable access is contingent on infrastructural investment and community values. By interpreting the Delphi findings through this ecosystemic lens, the implications of AI integration become clearer: sustainable and ethical adoption cannot be achieved through piecemeal interventions but requires coordinated action across policy, practice, and research domains. This ecosystemic interpretation also clarifies the paper's relationship to recent global frameworks, particularly UNESCO's guidance on generative AI in education and research [34]. Rather than treating competence, governance, transparency, teacher development, and equity as separate policy concerns, the Delphi findings suggest that these dimensions are mutually dependent within AI-enhanced educational ecosystems. In this respect, the study complements UNESCO's guidance by providing empirical evidence that the conditions identified in global policy discourse are also those that experts across research, policy, and practice prioritise most strongly, while revealing where implementation pressures are perceived to be greatest.

Contextual complexities and structural tensions while integrating AI in educational ecosystems

A critically reflective reading of these findings, however, demands that we move beyond cataloguing priorities and attend more rigorously to the contextual dynamics and inherent tensions that surface when AI integration is pursued within already complex, institutionally embedded activity systems. Educational institutions are not neutral containers into which new technologies can be straightforwardly inserted; they are sociotechnical systems shaped by historical practices, professional cultures, power relations, and competing stakeholder interests [26]. The five trends identified in this study do not exist in isolation from one another, nor from these broader systemic forces, and the tensions between them deserve explicit scholarly attention.

One of the most consequential tensions revealed by the data concerns the relationship between pedagogical transformation and teacher professional identity. The evolved roles of teachers working alongside AI agents received one of the highest challenge ratings in the entire dataset ($M = 8.40$), yielding the only negative priority indicator among the top-ranked themes. This finding is analytically significant: it suggests that even as experts broadly endorse the transformative potential of AI-mediated teaching, they simultaneously recognise that the reconfiguration of professional roles is a deeply contested terrain. Teachers are not passive recipients of technological change; they are professionals whose identities, pedagogical beliefs, and institutional loyalties actively mediate how, whether, and on what terms they engage with AI tools [2, 4, 44]. The risk, therefore, is that top-down mandates for AI integration—without adequate attention to the experiential and identity-related dimensions of teaching—may generate resistance, superficial compliance, or professional disengagement, all of which would undermine the long-term sustainability of educational transformation.

A second, closely related tension arises between the imperative of competence-building and the structural conditions in which such development must occur. Although AI-related competences for learners and educators emerged as the highest-rated theme in both impact ($M = 8.57$) and importance ($M = 8.64$), the assumption that competence development can be treated as a discrete, solvable problem obscures deeper systemic challenges. Professional learning opportunities are unevenly distributed across educational systems, with disparities linked to institutional resources, geographic location, socioeconomic context, and teacher workload [23, 40]. Moreover, competence frameworks for AI in education remain underspecified and contested, reflecting broader disagreements about what it means to be AI-literate in pedagogically meaningful terms, rather than merely technically functional ones [9, 19]. Future research must therefore address not only what competences are needed, but how structural inequalities within educational ecosystems constrain or enable their development across diverse institutional and geopolitical contexts.

A third tension emerges between the promise of automated formative assessment and the sociotechnical preconditions necessary to realise it responsibly. The expert panel rated automated assessment as highly impactful ($M = 8.13$) while acknowledging moderate but non-trivial challenges ($M = 7.13$). What the quantitative data alone cannot fully convey is the profound sociotechnical complexity involved in deploying AI-driven assessment systems in ways that are pedagogically valid, culturally sensitive, and equitable. Automated feedback systems are trained on historically produced datasets that embed existing biases, and their outputs can differentially advantage or disadvantage learners depending on linguistic background, learning style, and socioeconomic context [6, 37]. Furthermore, the institutional integration of such systems is seldom straightforward: it requires alignment with curriculum standards, assessment policies, data governance frameworks, and the professional judgements of individual educators—each of which may pull in different directions [39]. Treating automated assessment as primarily a technical problem, rather than as a sociotechnical and pedagogical one, risks replicating or amplifying existing educational

inequalities under a veneer of algorithmic objectivity.

The themes of explainability, trustworthiness, and equitable access reveal a fourth, and arguably the most structurally pervasive, tension in the data: the potential for AI integration to exacerbate rather than ameliorate educational inequality. Both themes received high importance ratings ($M = 8.39$ and $M = 8.19$, respectively) alongside high challenge scores ($M = 8.17$ and $M = 8.00$), and their low priority indicators signal that the gap between aspiration and implementation is particularly wide. Access to AI-enhanced educational tools is contingent upon infrastructure, connectivity, device availability, and digital fluency—resources that are systematically unequal across and within national educational systems [3,6]. Without deliberate, equity-oriented policy intervention, the diffusion of AI in education may follow the well-documented pattern of educational technology adoption, in which well-resourced institutions and learners capture disproportionate benefits while already-marginalised communities are further left behind [13,22]. This dynamic calls for a critical ecosystemic perspective that explicitly interrogates how the structural conditions of educational systems—funding regimes, regulatory frameworks, labour relations, and community values—shape the distributional outcomes of AI adoption [27,28].

Taken together, these tensions underscore that AI integration in educational ecosystems is not merely a technical or pedagogical challenge but a fundamentally political and ethical one. Sustainable, long-term educational transformation requires that policymakers, researchers, and practitioners engage explicitly with the contextual dynamics and structural contradictions that the present study has surfaced, resisting reductive narratives of technological inevitability and attending instead to the complex, contested processes through which new sociotechnical arrangements become embedded in—or rejected by—educational institutions and communities. Future research would benefit from adopting longitudinal, ethnographic, and participatory methodologies to track how these tensions evolve across diverse institutional contexts, and from developing theoretically grounded frameworks to guide equitable, adaptive, and democratically accountable AI integration over time [21,26].

Strategies and actions

Based on the findings of the Delphi study and the persona reflections, as well as informed by current research findings, we recommend the following actions for policy makers (PM), researchers (RE) and practitioners (PR), with each strategy linked to the corresponding challenges identified above:

- In order to foster a deep understanding of how education ecosystems are shaped by—and actively contribute to—the broader societal landscape of artificial intelligence:

Facilitating exchange of knowledge and best practices between stakeholders inside and outside the education ecosystem (PM, RE, PR)

Fostering collaborative knowledge construction through equitable and sustainable public-private stakeholder partnerships (PM, RE, PR)

Creating an open culture that acknowledges and responds to the evolving needs, challenges and agency of stakeholders and the affordances of technology (PM, RE, PR)

- In order to understand and responsibly utilise multimodal data as a foundational component and product of AI-enhanced education ecosystems:

Create awareness, research and understand the effects of the bias of (synthetic and real-world) educational data and model performance (RE, PR, PM)

Utilise the principles of data science for co-designing AI-enhanced ecosystems with open data (RE, PR)

Formulate and implement actionable guidelines for equitable access, data protection, and privacy and use of data (PM)

- In order to ensure inclusive and equitable access to the AI-enhanced education ecosystems through accessible design, adequate infrastructure, and aligned policy frameworks:

Design and implement AI-enhanced education ecosystems guided by principles of inclusivity and equity (e.g. universal design for learning) (PR)

Invest and maintain in equitable infrastructure, connectivity, and support systems (PM)

Research, monitor and advise on the equity impacts of AI adoption and usage in the education ecosystem (RE)

- In order to establish and continuously reflect on a shared vision of a highly dynamic education ecosystem in relation to AI in education and society:

Openness towards new developments of AI-enhanced technology and infrastructure, including ongoing integration and evaluation (PM, RE, PR)

Accepting and shaping the evolving role of educators and pedagogies (PR, PM, RE)

- In order to iteratively develop, integrate, evaluate and implement acceptable usage of AI in the education ecosystem:

Synthesise, analyse, and utilise generic principles (e.g., risks and benefits of adoption, intellectual property) for acceptable AI in the education ecosystem (PR)

Identify, understand and adhere to generic principles for acceptable, trustworthy, transparent and unbiased use of AI in educational ecosystems (PR, PM, RE)

Conduct interdisciplinary research focused on the impacts of AI-enhanced educational ecosystems and develop new principles for acceptable usage (RE)

- In order to foster a culture of lifelong learning in the context of quickly evolving developments of AI in the education ecosystem:

Iteratively identify, review and facilitate the development of knowledge, skills and competencies needed for an AI-enhanced society (PR, RE, PM)

Provide ongoing and seamless professional learning opportunities in formal, non-formal and informal settings (PR)

Identify and allocate the associated resources (e.g. time, budget, people) to support a culture of lifelong learning (PM)

Research, monitor and advise on the impact of AI on teaching, learning, model performance, socio-emotional learning and degeneration of knowledge, etc. (RE)

Conclusion

This study engaged experts from research, industry, and policy through a Delphi process to identify critical priorities that must shape the next phase of AI adoption in education. The findings point to a future in which AI-related competencies for both learners and educators become foundational, enabling all stakeholders to participate confidently and critically in AI-supported learning environments. Experts emphasised the growing need for automated formative assessment and AI-driven feedback, as well as the evolving role of teachers who must increasingly orchestrate learning alongside AI agents, rather than merely using traditional tools. At the same time, the study highlights an urgent need to ensure Explainability and Trustworthiness in AI systems to maintain transparency, safeguard learner autonomy, and cultivate informed human-AI collaboration. Equally vital is the commitment to accessibility and equitable access, ensuring that the benefits of AI are distributed fairly across socioeconomic, geographic, and linguistic divides. Collectively, these priorities signal the emergence of a collaborative future where AI serves as an enabler, supporting educators, empowering learners, and strengthening educational ecosystems through human-centred, responsible innovation.

CRediT authorship contribution statement

Rwitajit Majumdar: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dirk Ifenthaler:** Writing – review & editing, Writing – original draft, Resources, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Clara Schumacher:** Writing – review & editing, Writing – original draft, Investigation. **Enda Donlon:** Writing – review & editing, Writing – original draft, Investigation. **Hsiao-Ping Hsu:** Writing – original draft, Investigation. **Jason Zagami:** Writing – original draft, Investigation. **Maaïke Heitink:** Writing – original draft, Investigation. **Wolfgang Mueller:** Writing – original draft, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1: 56 Trends collected from the experts that were compiled as 15 themes

1. AI agents as Virtual Teaching Assistants
 - 1.1. Virtual teaching assistants
 - 1.2. Intelligent Tutoring Systems using AI
2. AI-enhanced Educational Analytics
 - 2.1. AI-Enhanced Learning Analytics
 - 2.2. AI-Enhanced Learning Analytics - implications for teachers/educators and students
 - 2.3. Interplay of Generative AI and Learning Analytics
 - 2.4. Authorable Learning Analytics Dashboards combined with Generative AI
 - 2.5. Predictive Analytics for Student Performance
3. AI-assisted Educational Content Creation
 - 3.1. AI and/in Content Creation - potentials and problems for education (e.g. 'AI Slop', inaccurate/unreliable content, hallucinations, 'baked in' bias, etc.)
 - 3.2. Generative AI for creation of educational materials/contents
 - 3.3. AI for Creating Educational Materials
 - 3.4. Emergence of Natural Language Programming for Educational Applications
4. AI for Educational Administrative Efficiency
 - 4.1. AI and surveillance (monitoring user/student engagement across multiple and large data sets, etc)
 - 4.2. AI for administrative efficiency
 - 4.3. Automation of Teacher Tasks
5. AI Augmentation with Educational Edge Devices
 - 5.1. Seamless Integration of AI with Augmented and Virtual Reality (AR/VR)
 - 5.2. AI-Enhanced Gamification in Learning
 - 5.3. Personal Android Robots on Education: Reframing Learning and Pedagogical Models
6. Human-AI Collaborative models in Learning Processes
 - 6.1. AI facilitated behavioural modification at societal scale through education, mass and social media
 - 6.2. Human-AI Collaboration in Learning Processes
 - 6.3. Blended Human-AI Learning Ecosystems
7. Interdisciplinary Curriculum through AI-Driven Knowledge Graphs
 - 7.1. Optimizing STEM-Humanities Interdisciplinary Curriculum through AI-Driven Knowledge Graphs
 - 7.2. Reframing curriculum from collective to individual learning pathways and outcomes.
 - 7.3. AI in Online and Blended Learning
8. AI-Powered transformations: Data wrappers and middle wares
 - 8.1. AI-Powered Data Wrappers: Enhancing Institutional Data Insights without Disrupting Legacy Systems
 - 8.2. AI-Augmented Middleware Solutions: Enabling Interoperability in Outdated Educational Data Systems
 - 8.3. Intelligent Integration with Legacy Systems
 - 8.4. Federated and Distributed AI Models for Collaborative Innovation
9. Personalized and Adaptive learning with AI agents
 - 9.1. Personalised (adaptive) learning
 - 9.2. AI for personalised/adaptive learning
 - 9.3. Generative AI for personalized and adaptive learning (e.g., learning companions)
 - 9.4. Analyzing, predicting and generating individualized content and learning pathways for students that are not biased and transparent to the humans.
 - 9.5. Personalisation of AI through digital twinning, constructing aspirational student models to guide learning processes.
 - 9.6. Adaptive learning systems using AI
 - 9.7. Personalized Learning with GenAI
 - 9.8. Scaling Big Data and Learning Analytics for Real-Time Personalised Learning Support
10. Automated assessment and feedback with AI agents
 - 10.1. Automated assessment and feedback using LLMs
 - 10.2. Generative AI in assessments: adjustment of tasks, use for automated grading (unstructured data), development of competence models, provision of feedback
 - 10.3. Interactive AI chatbots for automated assessment and feedback systems
 - 10.4. The potential of AI for adaptive/formative feedback
 - 10.5. Reframing assessment from knowledge retrieval and procedural application to dispositional thinking skills demonstrating expert approaches.
 - 10.6. Natural Language Processing (NLP) in assessment
11. AI-related competences for learners and educators
 - 11.1. Data Literacy for teachers and students in the age of AI
 - 11.2. AI-related competences for learners and educators
 - 11.3. Using the human change process and adoption cycle to smooth system-wide implementation of AI and decrease automation anxiety and resistance
 - 11.4. Training in using AI to foster learning (for students and teachers)
12. AI-assisted skill development
 - 12.1. Using big data on career and job opportunities to craft multiple pathways for students
 - 12.2. AI for the development of (21st century) skills
 - 12.3. The Impact of AI-Assisted Writing on Adolescent Cognitive Development
13. Evolved roles of teachers while teaching with AI agents
 - 13.1. Generating research-based evidence-proven instructional strategies for educators to use with students
 - 13.2. AI for Teachers
 - 13.3. Role change of teachers
14. Explainability and Trustworthiness of AI agents
 - 14.1. Trustworthiness of the algorithms
 - 14.2. Explainable AI (xAI) in Education:
15. Accessibility and Equal Access of AI agents
 - 15.1. AI Integration in Schools
 - 15.2. Accessibility and Equal Access
 - 15.3. Accessibility

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